Data Extrapolation From Learned Prior Images for Truncation Correction in Computed Tomography

Yixing Huang, Alexander Preuhs, Michael Manhart, Guenter Lauritsch, and Andreas Maier, *Senior Member, IEEE*

***Abstract*— Data truncation is a common problem in computed tomography (CT). Truncation causes cupping artifacts inside the field-of-view (FOV) and anatomical structures missing outside the FOV. Deep learning has achieved impressive results in CT reconstruction from limited data. However, its robustness is still a concern for clinical applications. Although the image quality of learning-based compensation schemes may be inadequate for clinical diagnosis, they can provide prior information for more accurate extrapolation than conventional heuristic extrapolation methods. With extrapolated projection, a conventional image reconstruction algorithm can be applied to obtain a final reconstruction. In this work, a general plug-and-play (PnP) method for truncation correction is proposed based on this idea, where various deep learning methods and conventional reconstruction algorithms can be plugged in. Such a PnP method integrates data consistency for measured data and learned prior image information for truncated data. This shows to have better robustness and interpretability than deep learning only. To demonstrate the efficacy of the proposed PnP method, two state-of-the-art deep learning methods, FBPConvNet and Pix2pixGAN, are investigated for truncation correction in cone-beam CT in noise-free and noisy cases. Their robustness is evaluated by showing false negative and false positive lesion cases. With our proposed PnP method, false lesion structures are corrected for both deep learning methods. For FBPConvNet, the root-mean-square error (RMSE) inside the FOV can be improved from 92 HU to around 30 HU by PnP in the noisy case. Pix2pixGAN solely achieves better image quality than FBPConvNet solely for truncation correction in general. PnP further improves the RMSE inside the FOV from 42 HU to around 27 HU for Pix2pixGAN. The efficacy of PnP is also demonstrated on real clinical head data.**

***Index Terms*— Deep learning, robustness, interpretability, truncation correction, computed tomography.**

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Yixing Huang, Alexander Preuhs, and Andreas Maier are with the Pattern Recognition Lab, Friedrich-Alexander-University Erlangen-Nuremberg, 91054 Erlangen, Germany (e-mail: yixing.yh.huang@fau.de; alexander.preuhs@fau.de; andreas.maier@fau.de).

Michael Manhart and Guenter Lauritsch are with the Siemens Healthcare GmbH, 91052 Erlangen, Germany (e-mail: michael.manhart@siemens-healthineers.com; guenter.lauritsch@siemens-healthineers.com).

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I. Introduction

CONE-BEAM computed tomography (CBCT) with flat-panel detectors is widely used in interventional applications since in many interventions both cross-sectional imaging and real time fluoroscopy are required. Data truncation is a common problem in CBCT. It mainly arises in two scenarios:

a) In some clinical applications, only a certain region of the patient is of interest. To reduce unnecessary X-ray dose exposure to the patient, collimators are placed between the X-ray source and the detector for region-of-interest (ROI) reconstruction, which is also called interior tomography.

b) Due to the limited size of detectors, the body of large patients or external fiducial markers placed on patients cannot be entirely positioned inside the field-of-view (FOV). Transferring such patients from an interventional suite to a diagnostic suite delays intervention procedures. Meanwhile, it significantly impacts the diagnostic CT workflow. In busy radiology departments, this can decrease patient throughput where procedural complications can lead to significant scheduling conflicts [1]. Therefore, switching to a full scan is not favored. In addition, mounting larger detectors demands high requirement on hardware configuration, especially on X-ray tubes for large cone-angle coverage. As CBCT systems are typically for multiple purposes, the current detector size is sufficient for most other intervention applications. Therefore, mounting larger detectors is not favored either.

Images reconstructed from truncated data using the standard filtered back-projection (FBP) algorithm suffer from cupping artifacts inside the FOV, while anatomical structures outside the FOV are missing or severely distorted. Therefore, truncation correction stands for cupping artifact reduction inside the FOV and anatomical structure restoration outside the FOV in this work. For truncation correction, reducing cupping artifacts to improve image quality inside the FOV is the primary goal, since the region inside the FOV is the target region for diagnosis or interventional surgeries. The secondary goal is to extend the FOV. FOV extension has clinical value in the following aspects:

a) In interior tomography, due to the small size of ROI, it is not straightforward to get the relative position of structures of interest with respect to others, although their absolute position

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is known. Hence, the approximate reconstruction of peripheral anatomical structures is beneficial for locating structures of interest.

b) FOV extension provides beneficial information of anatomical structures or external fiducial markers outside the FOV, which is necessary for certain image-guided therapies. For example, in spine surgeries [2], fiducial markers, which are placed outside the FOV and reconstructed from truncated projections, are distorted, which impairs the identification accuracy. Therefore, image quality improvement outside the FOV is beneficial to improve marker detection accuracy.

To deal with data truncation, data extrapolation is the most straight-forward approach. For this purpose, heuristic extrapolation methods are widely applied, including symmetric mirroring [3], cosine function fitting [4], and water cylinder extrapolation (WCE) [5]. With such extrapolations, a smooth transition between measured and truncated areas is pursued to alleviate cupping artifacts inside the FOV.

As cupping artifacts are mainly caused by the global filtering step in the FBP algorithm, analytic reconstruction algorithms with alternative local filtering are proposed, for example, decomposing the ramp filter into a local Laplace filter and a nonlocal low-pass filter [6]. The discovery of differentiate back-projection (DBP) [7]–[10] is one milestone for interior tomography. With DBP, theoretically exact solutions can be developed based on *a priori* knowledge [11], [12].

With the advent of compressed sensing technologies, iterative reconstruction with total variation (TV) regularization became the mainstream for CT reconstruction from insufficient data [13]–[15]. For interior tomography where an ROI is piecewise constant or polynomial, the ROI can be exactly and stably reconstructed via TV minimization [16]–[18]. However, compressed sensing methods, as well as the above mentioned extrapolation methods and local filtering methods, are only able to improve the image quality inside the FOV, but fails to reconstruct anatomical structures outside the FOV accurately due to missing data.

Recently, deep learning has achieved impressive results in CT reconstruction [19], [20]. In the field of interior tomography, Han and Ye have proposed to use DBP reconstruction images instead of FBP reconstruction images as the input of the U-Net for better stability in various types of ROI reconstruction tasks [21]. In addition, interior tomography images can be directly learned from truncated data by the iCT-Net [22]. For FOV extension, Fournié *et al.* [23] have applied the U-Net to post-process images reconstructed from linear extrapolated data, which has already achieved improved image quality outside the FOV.

The above achievements have shown a promising prospect of the clinical application of deep learning in CT reconstruction. However, the robustness of deep learning in practice is still a concern [24]–[26]. It is known that deep learning cannot generalize well to unseen data [27], especially with insufficient training data. Meanwhile, it is reported that adversarial examples [28]–[30] are ubiquitous in deep neural networks. In medical applications, deep learning is sensitive to noise [31]. Due to the instability of deep learning, generating reconstructed images directly from a neural network appears inadequate since incorrect structures might occur in the learned images. However, images generated by deep learning methods can provide substantial beneficial information of anatomical structures. Therefore, applying learned images as prior is a trend in medical image reconstruction now [32]–[36]. Learned images provide an estimate of unmeasured data for image reconstruction from insufficient data, e.g., in limited angle tomography [32] and sparse-view CT [34]. In addition, they can also accelerate convergence in iterative reconstruction [35]. For truncation correction [33], more accurate data extrapolation can be achieved based on such learned images than simple approximations like water cylinders.

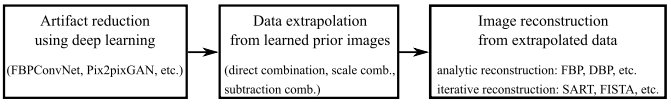


Fig. 1. A flowchart of our proposed PnP method for truncation correction. In the first and the last steps, various deep learning methods and conventional image reconstruction methods can be plugged in, respectively.

This work is an extension of our previous preliminary work [33]. The extension lies in the following aspects: a) Investigate the robustness of deep learning in truncation correction by showing false negative and false positive lesion cases; b) Extend the method into a general plug-and-play (PnP) form, where various deep learning methods and conventional image reconstruction methods can be plugged in; c) Apply adversarial learning to improve truncation correction; (d) The effects of different reconstruction algorithms are investigated.

II. Materials And Methods

As displayed in Fig. 1, our proposed PnP method mainly consists of three steps: artifact reduction using deep learning, data extrapolation from learned prior images, and image reconstruction from extrapolated data. In the first and the last steps, various deep learning methods and conventional image reconstruction methods can be plugged in, respectively.

A. Artifact Reduction Using Deep Learning

For CT image reconstruction, deep learning methods can be generally categorized to three types: a) sinogram inpainting in the projection domain (projection-to-projection translation) [37]–[41]; b) artifact post-processing in the image domain and image transformed domains (image-to-image translation) [21], [23], [42]–[49]; c) learning reconstruction images directly from projection data (projection-to-image reconstruction) [22], [50]–[57]. For projection-to-projection and image-to-image translation, incomplete sinograms and artifact corrupted images need to be translated to complete sinograms and artifact-free images, respectively. For projection-to-image reconstruction, known operators [58] for analytic or iterative reconstruction are typically integrated into the architecture design of deep neural networks. In cone-beam CT, due to the large amount of projection data, it is typically memory costly and computationally expensive for projection-to-projection and projection-to-image methods. Image-to-image methods typically adopt slice-wise or

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subvolume-wise training to overcome this problem. Therefore, in this work, we elaborate our PnP method using image-to-image translation methods for cone-beam CT truncation correction.

1) FBPConvNet: In medical imaging, the U-Net based methods are most widely used because of its concise architecture design, multi-scale feature extraction, fast convergence in training and high representation power. In this work, we choose the FBPConvNet [59] as the baseline method, where the U-Net is the backbone. The input of the FBPConvNet is the FBP (FDK in particular) reconstruction from WCE processed projections (denoted by $\boldsymbol{f}_{\mathrm{WCE}}$), while the output is the FBP reconstruction from nontruncated projection data. An ℓ_2 loss is applied for training. To reduce computational burden, we process the data slice-wise instead of (sub-)volume-wise. The correlation among different 2-D slices in the third dimension will be brought back in the third step of our PnP method.

2) Pix2pixGAN: For image-to-image translation, GANs are the state-of-the-art. Therefore, in this work, we propose to apply a pixel-to-pixel generative adversarial network (pix2pixGAN) [60] for truncation correction. The same U-Net in FBPConvNet is used as the generator G while a 5-layer CNN is used as the discriminator D [60]. G learns to convert a corrupted image to an artifact-free image. D learns to distinguish the output image from the target image. The objective of the conditional GAN is,

$$\begin{aligned}&\mathcal{L}_{\mathrm{cGAN}}(G, D)\\&\quad=\mathbb{E}_{\boldsymbol{x},\boldsymbol{y}}\left[\log D(\boldsymbol{x},\boldsymbol{y})\right]+\mathbb{E}_{\boldsymbol{x}}\left[\log\left(1-D(\boldsymbol{x},G(\boldsymbol{x}))\right)\right],\end{aligned}\tag{1}$$

where $\boldsymbol{x}$ is the input, $\boldsymbol{y}$ is the target, G tries to minimize this objective against an adversarial D that tries to maximize it, i.e., $G^*=\arg\min_G\max_D\mathcal{L}_{\mathrm{cGAN}}(G,D)$. In addition, the same ℓ_2 loss function in FBPConvNet is applied to train the generator's output close to the target,

$$\mathcal{L}_{\ell_2}=\mathbb{E}_{\boldsymbol{x},\boldsymbol{y}}\left[||(\boldsymbol{y}-G(\boldsymbol{x}))||_2\right].\tag{2}$$

The final objective for the Pix2pixGAN is

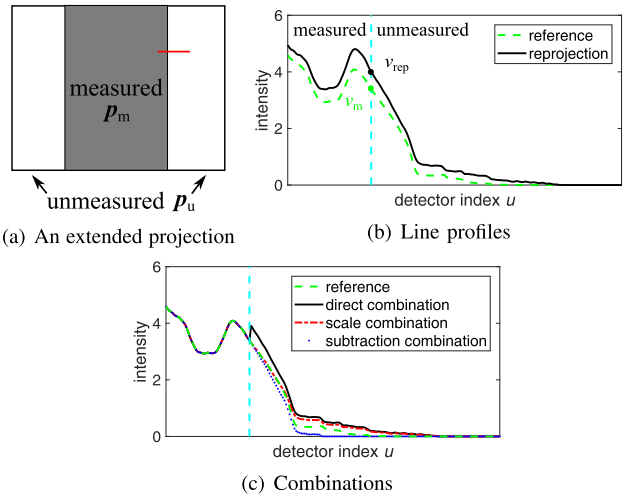
$$G^*=\arg\min_G\max_D\mathcal{L}_{\mathrm{cGAN}}(G,D)+\alpha\mathcal{L}_{\ell_2}.\tag{3}$$

B. Data Extrapolation From Learned Prior Images

Due to various factors such as overfitting, insufficient training data, or sensitivity to noise, the quality of images processed solely by neural networks may be inadequate. In such image-to-image translation methods, inference images have no direct connection to measured projection data. Hence, images with incorrect structures are likely not consistent with measured data. To further improve image quality, we propose to apply such learning-based images as prior for the extrapolation of truncated projection data, while the original measured projection data are preserved.

As displayed in Fig. 2(a), we denote measured projections by $\boldsymbol{p}_{\mathrm{m}}$ and their corresponding system matrix by $\boldsymbol{A}_{\mathrm{m}}$. We further denote unmeasured projections by $\boldsymbol{p}_{\mathrm{u}}$ and their corresponding system matrix by $\boldsymbol{A}_{\mathrm{u}}$. The learned reconstruction, denoted by $\boldsymbol{f}_{\mathrm{DL}}$, provides prior information for the unmeasured projections $\boldsymbol{p}_{\mathrm{u}}$. Therefore, an estimate of $\boldsymbol{p}_{\mathrm{u}}$, denoted by $\hat{\boldsymbol{p}}_{\mathrm{u}}$, is achieved using digitally rendered radiographs (DRRs) of $\boldsymbol{f}_{\mathrm{DL}}$,

$$\hat{\boldsymbol{p}}_{\mathrm{u}}=\boldsymbol{A}_{\mathrm{u}}\boldsymbol{f}_{\mathrm{DL}}.\tag{4}$$



(a) An extended projection (b) Line profiles

(c) Combinations

Fig. 2. Line profiles and different combination methods.

To get a complete projection set for reconstruction, one simple approach is to directly combine $\hat{\boldsymbol{p}}_{\mathrm{u}}$ with $\boldsymbol{p}_{\mathrm{m}}$. However, due to the error in $\boldsymbol{f}_{\mathrm{DL}}$, intensity discontinuity occurs between $\hat{\boldsymbol{p}}_{\mathrm{u}}$ and $\boldsymbol{p}_{\mathrm{m}}$ at the transition area. To illustrate this discontinuity, the intensity profiles of an example line (marked as the red line in Fig. 2(a) for the right side case) from the reference projection with an extended large detector and the corresponding line in the DDR $\hat{\boldsymbol{p}}_{\mathrm{u}}$ are displayed in Fig. 2(b), where the vertical dash line marks the truncation boundary. Typically there are deviations between these two line profiles. As a consequence, directly combining $\hat{\boldsymbol{p}}_{\mathrm{u}}$ with $\boldsymbol{p}_{\mathrm{m}}$ results in the black line profile in Fig. 2(c), where discontinuity is observed.

To reduce this intensity discontinuity in transition areas, we propose to modify the DDRs row-wisely according to the reprojected value v_{rep} and the measured value v_{m} on the truncation boundary pixel. Here we introduce two ways: scale combination (scaleComb) and subtraction combination (subtractComb). For scaleComb, scaling factors are computed as $s=v_{\mathrm{m}}/v_{\mathrm{rep}}$ row-wisely and afterwards the DDRs are rescaled by these scaling factors in the unmeasured part. For subtractComb, offset factors are computed as $b=v_{\mathrm{rep}}-v_{\mathrm{m}}$ and afterwards the DDRs are subtracted by these offset factors. Since projection values are non-negative, the negative values in subtractComb are set to zero. Fig. 2(c) shows an example of these two combination methods. The line profiles of both combination methods have lower deviation to the reference profile and more importantly they have smooth transition at the truncation boundary, compared with the direct combination (directComb). The modified DRRs are denoted by $\bar{\boldsymbol{p}}_{\mathrm{u}}$. Combining $\bar{\boldsymbol{p}}_{\mathrm{u}}$ with $\boldsymbol{p}_{\mathrm{m}}$, an estimate of the complete projection set with higher accuracy is obtained.

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C. Image Reconstruction From Extrapolated Data

With the above obtained complete projection set, analytic reconstruction algorithms such as FBP can be directly applied for image reconstruction. Analytic reconstruction is computationally efficient.

However, as indicated in Fig. 2(c), the extrapolated projection data $\bar{\boldsymbol{p}}_{\mathrm{u}}$ still have deviations to the reference. Therefore, the measured projections $\boldsymbol{p}_{\mathrm{m}}$ and the unmeasured ones $\bar{\boldsymbol{p}}_{\mathrm{u}}$ have different data fidelities. To incorporate different data fidelity terms, iterative reconstruction has advantages over analytic reconstruction as different weight or error tolerance parameters can be set respectively for measured and unmeasured projection data in the objective functions of iterative reconstruction.

1) Projection Domain Prior (PDP): With extrapolated projection data, images can be reconstructed according to the following general objective function, which consists of four parts: the data fidelity term $F_{\mathrm{m}}(\boldsymbol{f})$ for measured data, the data fidelity term $F_{\mathrm{u}}(\boldsymbol{f})$ for unmeasured data with its hyperparameter $\lambda \geq 0$, the regularization term R with its hyperparameter $\beta \geq 0$, and the indicator function $\iota_{\mathbb{R}_+}$,

$$L(\boldsymbol{f}) = F_{\mathrm{m}}(\boldsymbol{f}) + \lambda F_{\mathrm{u}}(\boldsymbol{f}) + \beta R(\boldsymbol{f}) + \iota_{\mathbb{R}_+}(\boldsymbol{f}). \tag{5}$$

The data fidelity for measured data is a weighted squared residual between the measured projection data and the forward projected reconstruction,

$$F_{\mathrm{m}}(\boldsymbol{f}) = ||\boldsymbol{\Lambda}_{\mathrm{m}}^{-1/2}(\boldsymbol{A}_{\mathrm{m}}\boldsymbol{f} - \boldsymbol{p}_{\mathrm{m}})||, \tag{6}$$

where $\boldsymbol{\Lambda}_{\mathrm{m}}$ is a diagonal matrix assigning a statistical weight to each measurement since noise typically exists in $\boldsymbol{p}_{\mathrm{m}}$. Depending on applications, different norms can be chosen for $||\cdot||$. Particularly in this work, we choose the ℓ_2 square norm $||\cdot||_2^2$. Similarly, the data fidelity for unmeasured data $F_{\mathrm{u}}(\boldsymbol{f})$ is defined as,

$$F_{\mathrm{u}}(\boldsymbol{f}) = ||\boldsymbol{\Lambda}_{\mathrm{u}}^{-1/2}(\boldsymbol{A}_{\mathrm{u}}\boldsymbol{f} - \bar{\boldsymbol{p}}_{\mathrm{u}})||, \tag{7}$$

where $\boldsymbol{\Lambda}_{\mathrm{u}}$ is another diagonal weight matrix.

The regularization term $R(\boldsymbol{f})$ mitigates noise or artifacts in the reconstruction and is defined in its general form,

$$R(\boldsymbol{f}) = \sum_{i=1}^{N}\sum_{j=1}^{N} c_{i,j}\,\psi(\boldsymbol{f}_i - \boldsymbol{f}_j), \tag{8}$$

where N is the number of voxels, i and j are voxel indices, $\boldsymbol{f}_i$ is the i-th voxel intensity, ψ is a potential function used to assign a cost to the difference between voxel values, and $c_{i,j} \in [0, 1]$ is a weight. For example, total variation regularization for $R(\boldsymbol{f})$ is commonly used in compressed sensing.

The indicator function $\iota_{\mathbb{R}_+}$ enforces non-negative voxel values for the reconstruction [61],

$$\iota_{\mathbb{R}_+} = \begin{cases} 0, & \text{if } \boldsymbol{f}_i \in \mathbb{R}_+ \text{ for } i = 1, \ldots, N, \\ +\infty, & \text{otherwise.} \end{cases} \tag{9}$$

By convention, the non-negativity constraint is applied in this work.

Since iterative reconstruction is computationally expensive, it is initialized by the deep learning reconstruction $\boldsymbol{f}_{\mathrm{DL}}$ to accelerate convergence,

$$\boldsymbol{f}^{(0)} = \boldsymbol{f}_{\mathrm{DL}}. \tag{10}$$

Eqn. (5) is an unconstrained optimization problem. Alternatively, the following constrained objective function can be used [13],

$$\min R(\boldsymbol{f}), \text{ subject to } \begin{cases} F_{\mathrm{m}}(\boldsymbol{f}) \leq e_{\mathrm{m}}, \\ F_{\mathrm{u}}(\boldsymbol{f}) \leq e_{\mathrm{u}}, \\ \boldsymbol{f}_i \geq 0 \text{ for } i = 1, \ldots, N, \end{cases} \tag{11}$$

where e_{m} is an error tolerance parameter for noise in the measured projection data and e_{u} is a tolerance parameter that accounts for the inaccuracy of the extrapolated projection data.

2) Image Domain Prior (IDP): The objective function in Eqn. (5) or Eqn. (11) fundamentally integrates prior information from learned images in the projection domain. It is worth noting that prior information from learned images can alternatively be applied in the image domain directly. To include IDP, the following objective function is applied,

$$L(\boldsymbol{f}) = F_{\mathrm{m}}(\boldsymbol{f}) + \lambda\Gamma(\boldsymbol{f}) + \beta R(\boldsymbol{f}) + \iota_{\mathbb{R}_+}(\boldsymbol{f}). \tag{12}$$

where $\Gamma(\boldsymbol{f})$ is a distance term to the learned prior image $\boldsymbol{f}_{\mathrm{DL}}$ with its hyperparameter $\gamma \geq 0$,

$$\Gamma(\boldsymbol{f}) = ||\boldsymbol{f} - \boldsymbol{f}_{\mathrm{DL}}||. \tag{13}$$

In the learned images $\boldsymbol{f}_{\mathrm{DL}}$, error exists not only outside the FOV, but also inside the FOV. In some cases, the error inside the FOV is large (see Fig. 4(j) as an example). The IDP objective function of Eqn. (12) finds an image close to $\boldsymbol{f}_{\mathrm{DL}}$. As a consequence, the large error inside the FOV of $\boldsymbol{f}_{\mathrm{DL}}$ is partially preserved by the distance term $\gamma\,\Gamma(\boldsymbol{f})$, especially when γ is large. Instead, in the PDP objective function in Eqn. (5) or Eqn. (11), the learned prior image $\boldsymbol{f}_{\mathrm{DL}}$ contributes to the unmeasured projection data only, corresponding to the area outside the FOV. Therefore, the large error inside the FOV can be corrected by the data consistency with measured data. Thus, the PDP objective function in Eqn. (5) or Eqn. (11) is expected to be superior.

D. Experimental Setup

1) Experiments on AAPM Data: In this work, the proposed PnP method is first evaluated on AAPM CT data [62] in a simulated cone-beam CT system without and with noise. The simulated cone-beam CT system parameters are aligned with a Siemens Artis zee angiographic C-arm system. The source-to-detector distance is 1200 mm and the source-to-isocenter distance is 750 mm. The detector size is 1240 × 960 pixels with an isotropic detector pixel size of 0.308 mm. The system has an FOV diameter of 23.9 cm. The regular scan mode acquires 496 projections in an angular range of 200° with an angular increment around 0.4°. The fast scan mode acquires 248 projections in an angular range of 200° with an angular increment of 0.8° using a 2 × 2 rebinned detector. In the simulation experiments, the fast scan mode is used. The reconstruction volume size is 256 × 256 × 256 with a voxel size of 1.25 mm × 1.25 mm × 1.0 mm. We choose the fast mode and this coarse reconstruction resolution to save computation in cross-validation experiments, but it is sufficient for a proof-of-concept study. To generate DRRs for data extrapolation, a ray-driven forward projection method

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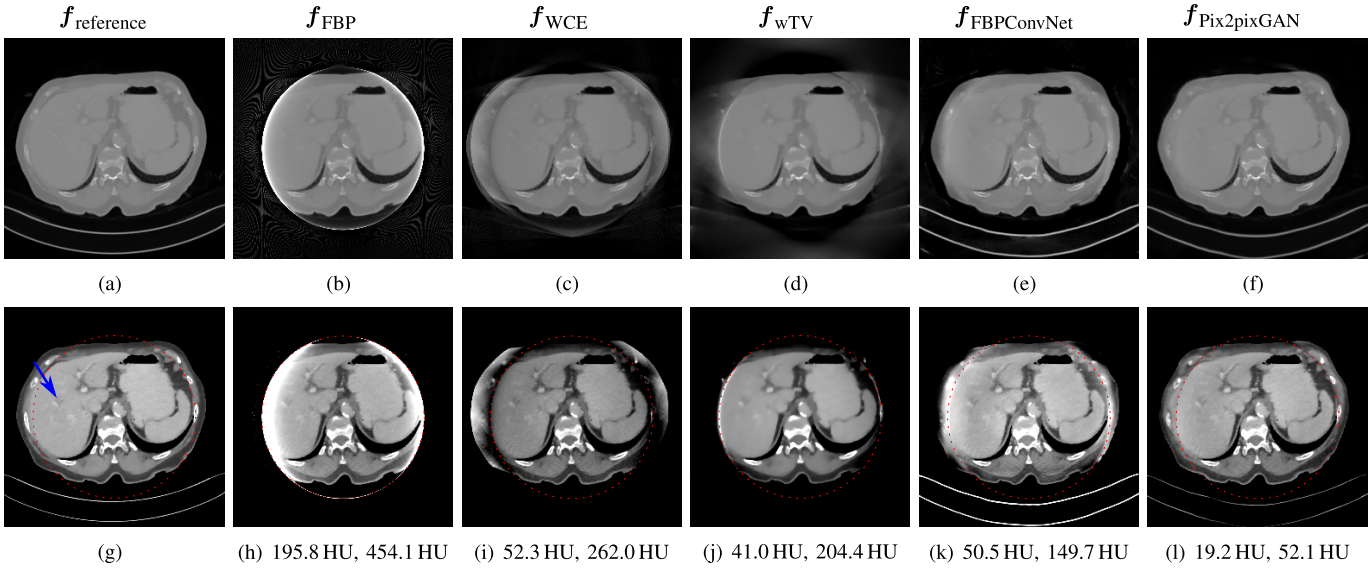


Fig. 3. Results of one example test slice in the noise-free case displayed in window [−1000, 1000] HU (top) and window [−200, 200] HU (bottom). The FOV boundary is marked by the dotted circle in the bottom images. The RMSE values inside (left) and outside (right) the FOV are displayed at the bottom subcaptions.

with a sampling rate of 7.5 samples per mm is applied. For all experiments considering noise, Poisson noise is simulated assuming an exposure of $I_0 = 10^5$ photons at each detector pixel before attenuation.

For deep learning, 18 patients' CT data sets are used, with a split of 16-1-1 for training, validation and test, respectively. To investigate the performance of deep learning on different training and test data, leave-one-out cross-validation is performed among 17 patients' data sets, while one patient is always used for validation. For training, 25 slices from each patient are chosen with a neighbouring slice distance of 10 mm. For validation, the root-mean-square error (RMSE) of 25 slices is used for monitoring the training process to avoid over-fitting. For noise-free evaluation, the neural network models are trained on noise-free data, while they are trained on data with the same level of Poisson noise in the noisy cases. The FBPConvNet and Pix2pixGAN are trained on the above data for 300 epochs. The initial learning rate is 10^{-3} and the decay rate is 0.97 for each epoch. For Pix2pixGAN, the weight parameter α is set to 100 in Eqn. (3). For the test patient, all the slices are fed to the trained model for evaluation.

In the noisy case, measured projections are preprocessed by projection domain PWLS according to [63], where the weight Λ_{m} is updated iteratively using 5 iterations. The weight Λ_{u} for unmeasured data is simply set to an identity matrix. For iterative reconstruction, we compare the difference between PDP and IDP objective functions (Eqn. (5) and Eqn. (12)) with parameters $\lambda = 0.5, \beta = 0$ and $\lambda = 0.1, \beta = 0$ empirically. To investigate whether the regularization term $R(\boldsymbol{f})$ is necessary, the results of PDP with $\lambda = 0.1, \beta = 0.1$, where iterative reweighted total variation (wTV) [15] is chosen for $R(\boldsymbol{f})$, are compared with those of $\lambda = 0.1, \beta = 0$. The objective functions are optimized by a fast iterative shrinkage-thresholding algorithm (FISTA) with 10 iterations.

To assess image quality, the metrics of RMSE and structure similarity (SSIM) are utilized. Since the patient beds are not of interest, they are excluded for image quality assessment by excluding the bottom 66 rows approximately. Since lesions typically have small areas, whether they exist or not cannot make significant changes on the standard image quality metrics (e.g., RMSE and SSIM). Therefore, false negative and false positive lesion cases are examplified to highlight the clinical value of our proposed PnP algorithm.

2) Experiment on Clinical Head Data: The Pix2pixGAN-based methods are also evaluated on a 3-D clinical head dataset with a typical noise level acquired from an Artis zee angiographic C-arm system (Siemens Healthcare GmbH, Forchheim, Germany). The dose area product (DAP) of the complete scan is 532 μ Gy·m^2. The detector size is 1240 × 960 pixels with an isotropic detector pixel size of 0.308 mm. The complete dataset contains 496 projections obtained in a 200° short scan. The reconstructed volume has a size of 512 × 512 × 300 with an isotropic voxel size of 0.465 mm. For the truncation setting, the projection data at the central 600 × 960 pixels are used. The FOV diameter is decreased from 23.9 cm to 11.6 cm.

To investigate the influence of training data size, two new Pix2pixGAN models are trained using 1430 slices selected from 53 patients and 2880 slices from 115 patients respectively in the CQ500 head CT database [64]. To get a similar level of noise, Poisson noise is added into the input training slices considering an exposure of $I_0 = 5 \times 10^5$ photons at each detector pixel before attenuation. The neural network training parameters and the reconstruction parameters are the same as those for the AAPM data.

III. Results

A. Results on AAPM Data in the Noise-Free Case

1) Noise-Free Case: The reconstruction results of one example test slice in the noise-free case are displayed in Fig. 3. The top row images are shown in a wide window

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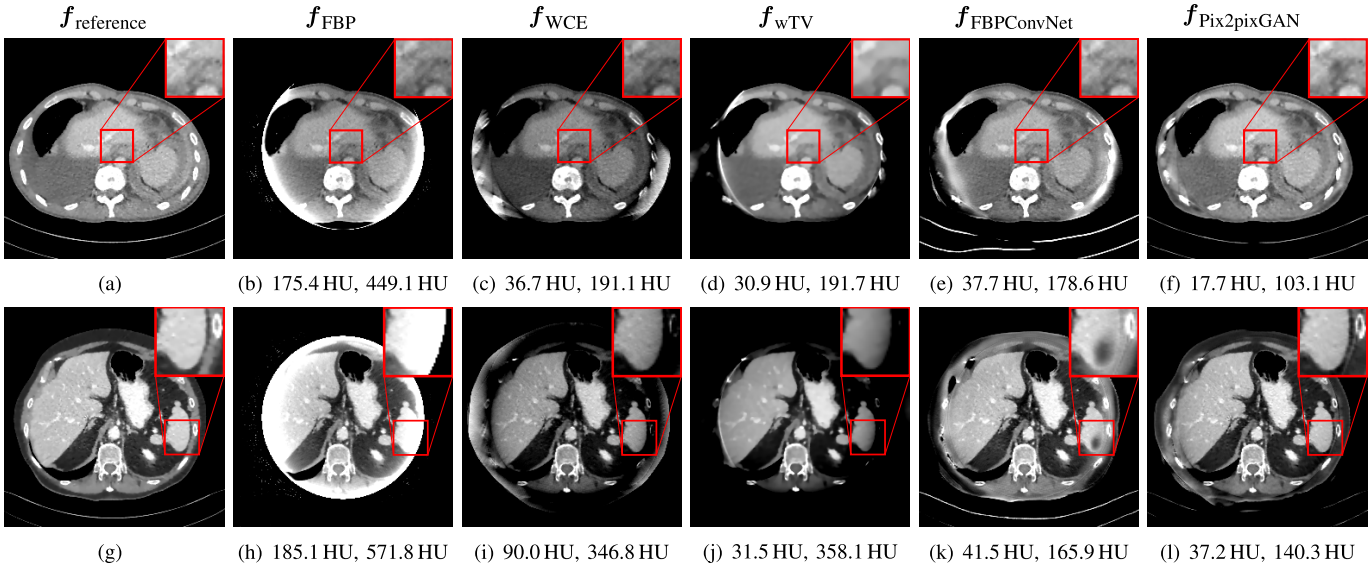


Fig. 4. False positive lesion examples in the ROIs of (f) and (k) reconstructed by Pix2pixGAN and FBPConvNet respectively in the noise-free case, window: [−200, 200] HU. The RMSE values inside (left) and outside (right) the FOV are displayed at the corresponding subcaption.

[−1000, 1000] HU. In the FBP reconstruction f_{FBP} in Fig. 3(b), the original FOV boundary is clearly observed. The structures inside the FOV suffer from cupping artifacts, while the anatomical structures outside this FOV are entirely missing. For f_{WCE} in Fig. 3(c), the cupping artifacts inside the FOV are notably alleviated by WCE. Anatomical structures outside the FOV are partially reconstructed. In the compressed sensing reconstruction f_{wTV} [15] (Fig. 3(d)), the cupping artifacts are removed. Nevertheless, the structures outside the FOV are still missing due to data truncation. Figs. 3(e) and (f) demonstrate that FBPConvNet and Pix2pixGAN are able to reduce the cupping artifacts as well. Moreover, they are able to reconstruct the anatomical structures outside the FOV, which demonstrates the efficacy of deep learning methods in FOV extension. Compared with Pix2pixGAN, FBPConvNet is slightly worse than Pix2pixGAN, as some areas have lower intensity than the reference, e.g., the area near the leftmost rib in Fig. 3(e).

In the bottom row, the images are redisplayed in a narrow window [−200, 200] HU. The RMSE values inside (left) and outside (right) the FOV are displayed at each corresponding subcaption. In this window, the detail structures of the liver, especially the lesion marked by the arrow in Fig. 3(g), can be observed. In the FBP reconstruction displayed in Fig. 4(h), the cupping artifacts are clearly present, which are mitigated by WCE, wTV, FBPConvNet and Pix2pixGAN. Among these reconstruction methods, Pix2pixGAN achieves the smallest RMSE values inside and outside the FOV, which is significant image quality improvement.

The lesion in Fig. 3(g) is reconstructed by FBPConvNet and Pix2pixGAN. However, not all lesions observed in the deep learning reconstruction are reliable, even in the noise-free case. Two example results are displayed in Fig. 4. In the ROI of Fig. 4(f), a lesion is located at the bottom tip of the liver, which looks very realistic. In Fig. 4(k), a large lesion is clearly visible. However, in the reference ROIs in Figs. 4(a) and (g), these two lesions do not exist. Therefore, false positive lesions might be reconstructed by FBPConvNet and Pix2pixGAN.

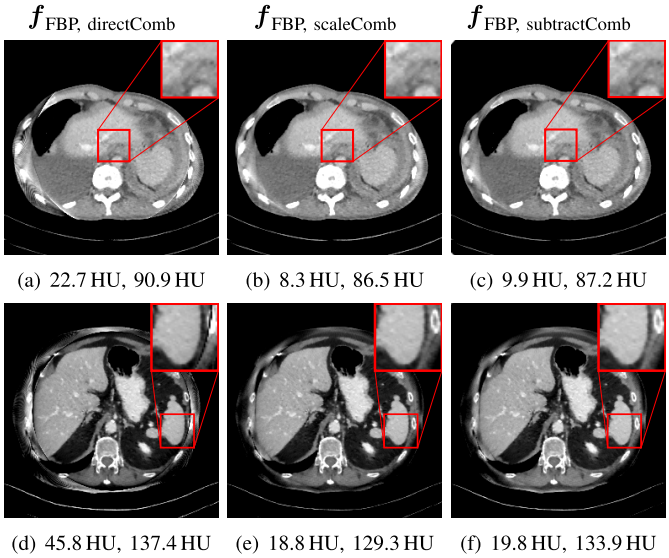
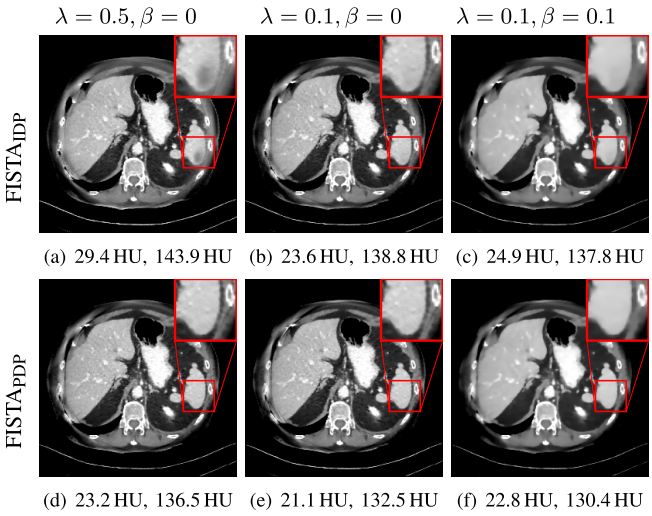


Fig. 5. The FBP reconstruction results with different extrapolation combination methods in the noise-free case, window: [−200, 200] HU. The top and bottom results use Fig. 4(f) and Fig. 4(k) as prior images for data extrapolation, respectively. The RMSE values inside (left) and outside (right) the FOV are displayed at the corresponding subcaption.

To avoid such incorrect structures inside the FOV, PnP applies the learned reconstruction images as prior for the extrapolation of truncated projection data. The FBP reconstruction results with different extrapolation methods for the two example test slices are displayed in Fig. 5. By preserving the original measured projection data, all the false positive lesions are corrected in Fig. 5. This demonstrates the efficacy of combining measured projection data with extrapolated data from learned images. However, with directComb, the circular FOV boundary is observed in Fig. 5(a) and (d).

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(a) 29.4 HU, 143.9 HU (b) 23.6 HU, 138.8 HU (c) 24.9 HU, 137.8 HU

(d) 23.2 HU, 136.5 HU (e) 21.1 HU, 132.5 HU (f) 22.8 HU, 130.4 HU

Fig. 6. The iterative reconstruction results solved by FISTA with different parameters in the noise-free case, window: [−200, 200] HU. All the results use Fig. 4(k) as prior for data extrapolation. The RMSE values inside (left) and outside (right) the FOV are displayed at the corresponding subcaption.

With scaleComb or subtractComb, this circular FOV boundary is reduced. For anatomical structures inside the FOV, no apparent difference is observed among these three methods.

The iterative reconstruction results using the IDP (Eqn. (12)) and PDP (Eqn. (5)) objective functions solved by FISTA, denoted by FISTA$_{IDP}$ and FISTA$_{PDP}$ respectively, with different parameters are displayed in Fig. 6. For IDP, when λ is set to 0.5, the false positive lesion in the FBPConvNet partially remains in the reconstruction result. When λ is set to 0.1, the false positive lesion is removed. Therefore, λ needs to be set to a relatively small value for IDP when large error is observed in the learned image. For PDP, when λ is set to 0.1 or 0.5, in both cases the false positive lesion is removed, indicating that PDP has more flexibility in parameter setting than IDP. To investigate whether additional regularization term is necessary, the results with wTV regularization [15] are displayed in Figs. 6(c) and (f). The corresponding methods are denoted by FISTA$_{IDP,\,wTV}$ and FISTA$_{PDP,\,wTV}$, respectively. The zoom-in ROIs indicate that wTV reduces some high frequency noise and fine structures as well. The RMSE values inside the FOV in the images with wTV are slightly higher than those without wTV, and the RMSE values outside the FOV have no significant improvement. They indicate that additional TV regularization is not necessary in the noise-free case.

The average RMSE and SSIM values of all the 17 test patients in the noise-free case are displayed in Tab. I. For different extrapolation methods from learned Pix2pixGAN images, both scaleComb and subtractComb achieve better image quality than directComb. Since the RMSE and SSIM values of scaleComb is slightly better than subtractComb, in the following experiments, we choose to use the scaleComb for demonstration. For Pix2pixGAN-based iterative reconstruction, different methods, independent of objective functions and whether using wTV regularization, achieve similar image quality, with average RMSE around 17 HU and 91 HU for the regions inside and outside the FOV, respectively. It also indicates that additional wTV regularization has no significant impact on restoring structures outside the FOV. In the following, we choose PDP (Eqn. (5)) for demonstration. Tab. I also shows that FBPConvNet solely achieves worse image quality than Pix2pixGAN solely. Nevertheless, FBPConvNet-based PnP results achieves comparable image quality as Pix2pixGAN-based PnP results inside the FOV.

TABLE I
THE QUANTITATIVE EVALUATION RESULTS IN THE NOISE-FREE CASE

Method	RMSE in FOV	RMSE outside FOV	SSIM in FOV	SSIM outside FOV
FBP	179.8 HU	417.9 HU	0.6015	0.0097
WCE	52.7 HU	229.6 HU	0.9856	0.7158
wTV	37.6 HU	183.4 HU	0.9941	0.7218
Pix2pixGAN	25.6 HU	101.7 HU	0.9977	0.9348
FBP$_{directComb}$	28.7 HU	90.0 HU	0.9955	0.9307
FBP$_{scaleComb}$	16.4 HU	85.3 HU	0.9991	0.9443
FBP$_{subtractComb}$	17.4 HU	86.5 HU	0.9989	0.9427
FISTA$_{IDP}$	17.7 HU	91.5 HU	0.9982	0.9461
FISTA$_{PDP}$	16.8 HU	91.9 HU	0.9987	0.9466
FISTA$_{IDP,\,wTV}$	20.5 HU	91.6 HU	0.9982	0.9464
FISTA$_{PDP,\,wTV}$	20.1 HU	90.1 HU	0.9983	0.9480
FBPConvNet	56.0 HU	187.9 HU	0.9849	0.7802
FBP$_{scaleComb}$	19.8 HU	126.6 HU	0.9983	0.8965
FISTA$_{PDP}$	21.0 HU	125.4 HU	0.9982	0.9074

B. Results on the AAPM Data in the Noisy Case

The truncation correction methods are also evaluated in the noisy case. The results of two example slices with FBPConvNet-based methods are displayed in Fig. 7. Noise patterns are observed in the WCE reconstruction (Figs. 7(b) and (h)), where the lesions in the corresponding ROIs are partially obscured by noise. Figs. 7(c) and (i) demonstrate that wTV is able to reduce Poisson noise very well and reveal the lesion areas. For the FBPConvNet result displayed in Fig. 7(d), the lesion boundary in the ROI is very blurry and its area is very difficult to determine. For the FBPConvNet result in Fig. 7(j), the low contrast (about 20 HU) lesion in the ROI is missing. These observations indicate that deep learning is not robust enough in the existence of noise, although the FBPConvNet is trained on noisy data. With our proposed PnP method, both lesions are reconstructed in Figs. 7(e)-(f) and (k)-(l). Since it is well-known that TV regularization is beneficial in noise reduction. Therefore, the wTV regularization term is applied in Figs. 7(f) and (l). Comparing Figs. 7(f) and (l) to Figs. 7(e) and (k) respectively, the noise in the zoom-in ROIs is reduced and the lesions are visualized better.

The results of the same two example slices with Pix2pixGAN-based methods are displayed in Fig. 8. More detailed anatomical structures are observed in the Pix2pixGAN results (Figs. 8(a) and (d)) than the FBPConvNet results (Fig. 7(d) and (j)). However, the lesion in Fig. 8(a) has a much smaller area size than that in the reference (Fig. 7(a)) and the low contrast lesion in Fig. 8(d) is missing. With our proposed PnP method, the lesions are reconstructed in Figs. 8(b)-(c) and (e)-(f).

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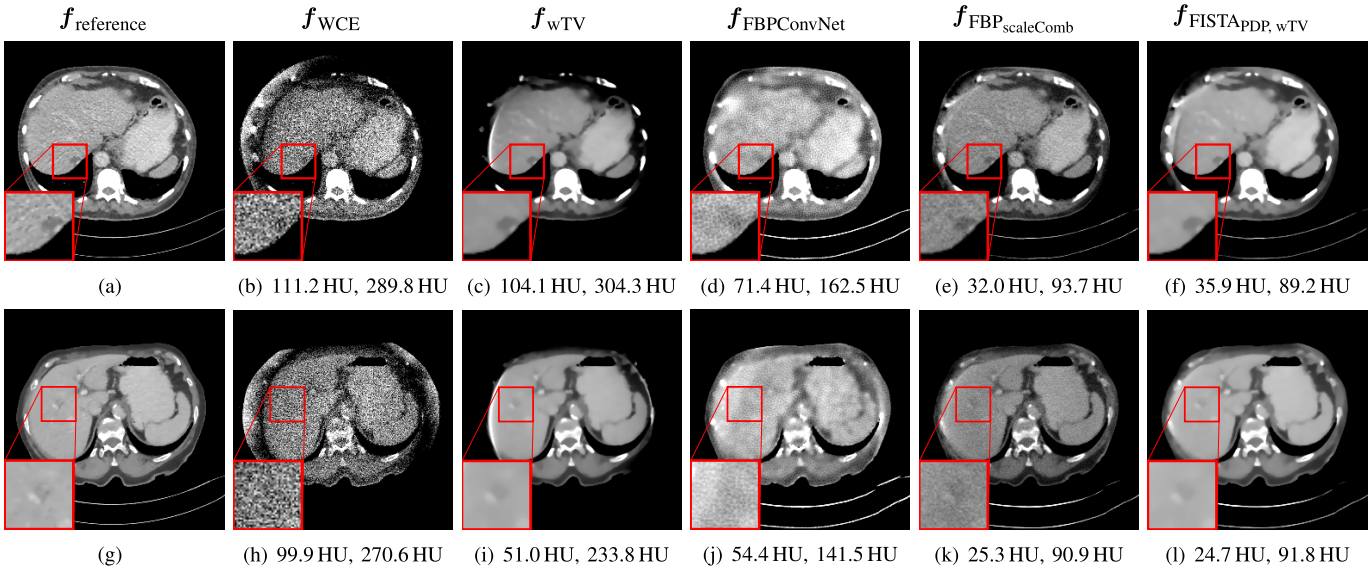


Fig. 7. Results of two example slices in the noisy case with FBPConvNet-based methods, window: [−200, 200] HU. The lesion boundary in the ROI of (d) is difficult to determine and the very low contrast lesion in the ROI of (j) disappears entirely. The RMSE values inside (left) and outside (right) the FOV are displayed at the corresponding subcaption.

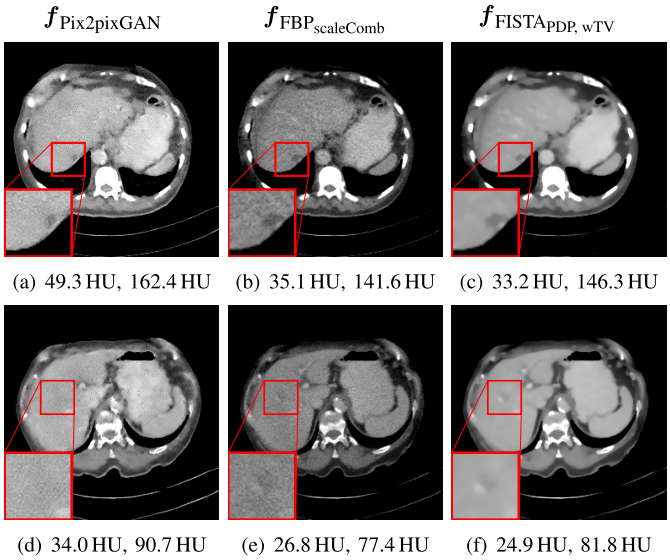


Fig. 8. Results of two example slices in the noisy case with FBPConvNet-based methods, window: [−200, 200] HU. Similar to the Pix2pixGAN results, the lesion in the ROI of (a) has incorrect outline and the very low contrast one in the ROI of (d) disappears entirely. The RMSE values inside (left) and outside (right) the FOV are displayed at the corresponding subcaption.

The quantitative evaluation results for all the test patients in the noisy case are displayed in Tab. II. It also indicates that FBPConvNet-based PnP results achieves comparable image quality as Pix2pixGAN-based PnP results inside the FOV, although FBPConvNet solely achieves worse image quality than Pix2pixGAN solely. Nevertheless, Pix2pixGAN-based PnP results have better image quality than FBPConvNet-based PnP results outside the FOV.

With the chosen parameters $\lambda = 0.1$ and $\beta = 0.1$ for PDP in the noisy case, the average RMSE values inside and outside the FOV of one test patient are plotted over iterations in Fig. 9.

TABLE II
THE QUANTITATIVE EVALUATION RESULTS IN THE NOISY CASE

Method	RMSE in FOV	RMSE outside FOV	SSIM in FOV	SSIM outside FOV
WCE	90.6 HU	235.7 HU	0.9543	0.7039
wTV	43.2 HU	225.8 HU	0.9842	0.6836
Pix2pixGAN	42.1 HU	122.9 HU	0.9920	0.9168
$\text{FBP}_{\text{scaleComb}}$	28.5 HU	103.4 HU	0.9968	0.9272
$\text{FISTA}_{\text{PDP, wTV}}$	26.2 HU	105.7 HU	0.9958	0.9364
FBPConvNet	91.9 HU	220.4 HU	0.9486	0.7633
$\text{FBP}_{\text{scaleComb}}$	31.7 HU	141.7 HU	0.9953	0.8751
$\text{FISTA}_{\text{PDP, wTV}}$	28.7 HU	140.6 HU	0.9944	0.8925

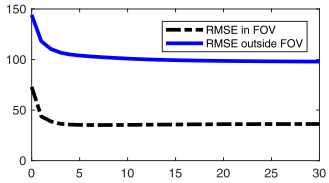


Fig. 9. The RMSE values inside and outside the FOV of one patient data over iterations using $\text{FISTA}_{\text{PDP, wTV}}$.

The RMSE inside the FOV converges faster than that of the whole image. However, both of them have little change after 10 iterations. Therefore, in this work, choosing 10 iterations for FISTA is sufficient.

C. Results on AAPM Data With an Extreme Prior Image

To demonstrate the property of our proposed PnP method in the existence of extremely incorrect learned prior images, the results of one test example from the AAPM data using a prior head reconstruction volume are displayed in Fig. 10. The reference images are displayed in Figs. 3(a) and (g). It is clear that the prior images in Figs. 10(a) and (d) are entirely different from the reference images. Nevertheless, with such

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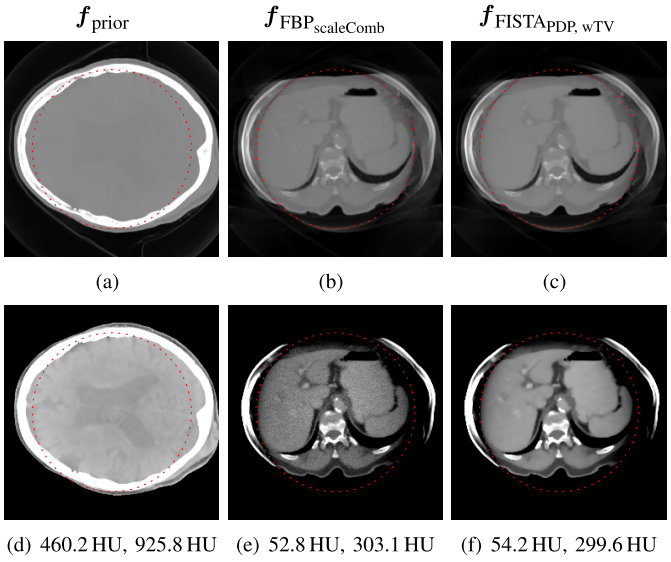


Fig. 10. Example results in the noisy case with an extreme prior image in window [−1000, 1000] HU (top) and window [−200, 200] HU (bottom). The reference images are displayed in Figs. 3(a) and (g), respectively. The RMSE values inside (left) and outside (right) the FOV are displayed at the bottom subcaptions.

a prior image for data extrapolation, our PnP method is still able to reconstruct anatomical structures inside the FOV well, although the structures outside the FOV are incorrect.

D. Pix2pixGAN-Based Results on Clinical Head Data

The results on the clinical head data are displayed in Fig. 11. Since there is no ground truth reconstruction, the FBP and the FISTA reconstructions from the complete PWLS processed data are displayed in Figs. 11(a) and (b) respectively for reference. In Fig. 11(a), streak artifacts (indicated by the arrow) are observed, which are potentially caused by the cone-beam angle, beam hardening, and scattering. In the WCE reconstruction (Fig. 11(c)), the anatomical structures outside the FOV are incorrect due to the large level of truncation. Hence, it is difficult to identify the relative location of the ROI in the head. In addition, streak artifacts are introduced due to the erroneous extrapolation, as displayed in the zoom-in ROI. The Pix2pixGAN result trained from 53 patients' data is displayed in Fig. 11(d). It reconstructs the anatomical structures outside the FOV, providing an overall anatomy outline, which is beneficial for location identification. The structures of the nose, the skull and the ears are reconstructed very well. The FBP reconstruction from scaleComb extrapolation in Fig. 11(e) has no apparent difference to Fig. 11(d). Nevertheless, the difference image in Fig. 11(h) indicates that $f_{\text{FBP}_{\text{scaleComb}}}$ has more accurate structures than $f_{\text{Pix2pixGAN}}$ with respect to the FBP reference in Fig. 11(a). The $\text{FISTA}_{\text{PDP,wTV}}$ reconstruction in Fig. 11(f) is able to reduce streak artifacts outside the FOV. In addition, the streak artifacts inside the FOV are reduced as well. Fig. 11(d)-(f) demonstrates the efficacy of PnP for real clinical applications. The Pix2pixGAN result trained from 115 patients' data is displayed in Fig. 11(j) and its difference image with respect to the FBP reference is displayed Fig. 11(m). With more training data, the peripheral soft tissues are reconstructed better in Fig. 11(j), e.g., the bottom soft tissue

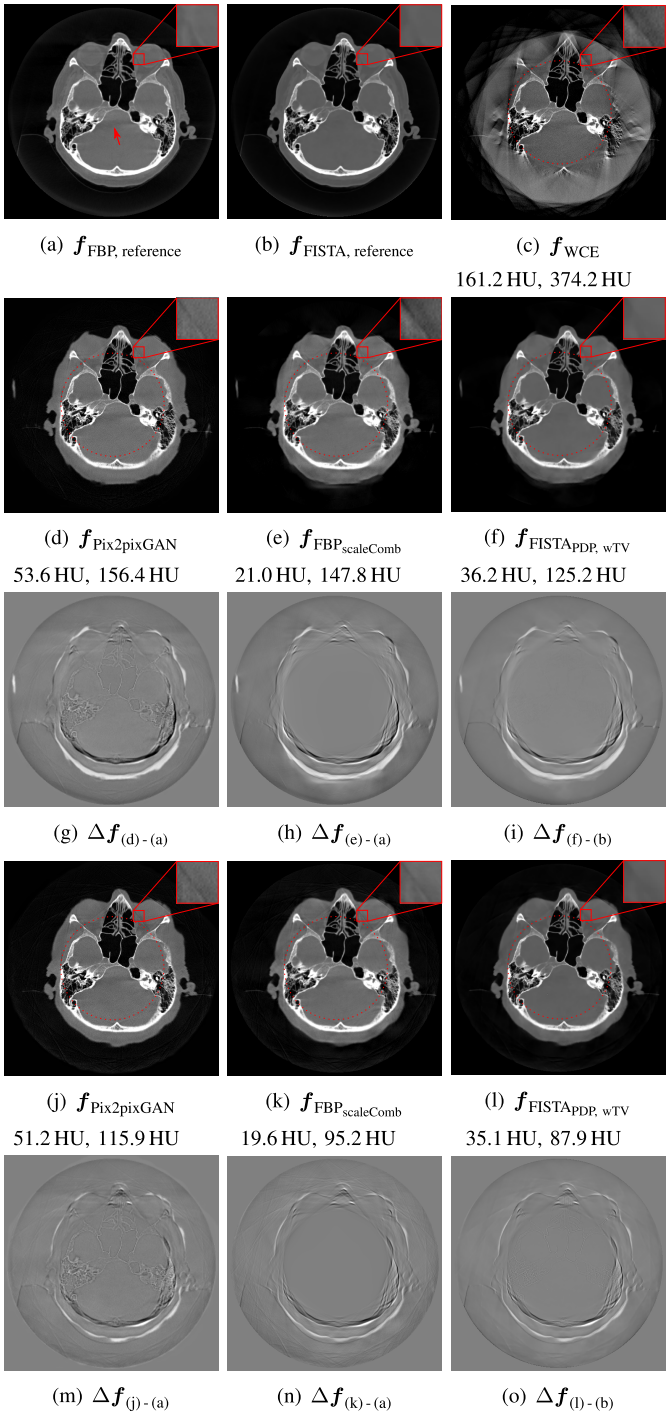


Fig. 11. The Pix2pixGAN-based results on the clinical head dataset, window: [−1000, 1000] HU. The results of (d)-(f) use 53 patients' data for training, while those of (j)-(l) use 115 patients' data. The difference images in (g)-(i) and (m)-(o) are displayed in a window width of 1000 HU. The RMSE values inside (left) and outside (right) the FOV are displayed at the corresponding subcaption.

boundary. As a result, the structures outside the FOV are also improved in the corresponding PnP results displayed in Fig. 11(k) and (l).

IV. Discussion

Deep learning methods achieve encouraging image reconstructions from truncated data. In a wide display window,

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high contrast structures (e.g., bones) and body outlines are typically reconstructed with high confidence, as displayed in Fig. 3(e) and (f). However, many fine details may be reconstructed incorrectly when they are displayed in a narrow window, although they appear very realistic. In this work, false negative lesion cases of deep learning methods are exhibited in Fig. 7(j) and Fig. 8(d). Moreover, false positive lesion cases of deep learning are discovered, as shown in Figs. 4(f) and (k). Especially, the lesion generated by Pix2pixGAN in Fig. 4(f) is so realistic that radiology experts may draw false diagnostic conclusions. Therefore, the observations in this work serve as a warning to the robustness of deep learning in clinical applications.

Noise is one main potential factor to influence the robustness of deep learning. The blurry structures in Figs. 7(d) and (j) demonstrate that FBPConvNet is sensitive to noise. In deep neural networks, as a consequence of high dimensional multiplications, noise will accumulate layer by layer and adds up to a large change to the output [28]. With adversarial learning, most anatomical structures in Figs. 8(a) and (d) are visualized well and Poisson noise is reduced to a large degree. Fig. 8 along with Tab. II indicates that Pix2pixGAN can reduce noise and truncation artifacts simultaneously better than FBPConvNet. Nevertheless, low contrast lesions might still disappear in Pix2pixGAN results (Fig. 8(d)).

Note that preprocessing (denoising) input images of neural networks is beneficial to improve the quality of deep learning results [65]. In this work, we feed the reconstruction from noisy projections directly to neural networks for three reasons: a) Demonstrate that Pix2pixGAN has better robustness than FBPConvNet to deal with noisy input images; b) Demonstrate that as long as learned images contain approximate body and organ outlines, although fine details are obscured by noise, they are able to provide more accurate information for extrapolation than conventional heuristic extrapolation methods like WCE; c) Demonstrate that the conventional reconstruction step in PnP is able to recover important anatomical structures inside the FOV for highly corrupted learned images.

Insufficient training data is another factor to influence the robustness of deep learning. The small training data size leads to the limited generalizability of deep learning algorithms to unseen data. Because of this, erroneous structures are reconstructed inside and outside the FOV in the noise-free case in this work, e.g., the erroneous structures in Fig.4. With our proposed PnP method, the erroneous structures inside the FOV are corrected in the third step by a conventional reconstruction algorithm. Therefore, PnP ensures the generalizability to unseen data inside the FOV. But the generalizability of PnP to unseen data outside the FOV is not guaranteed. With more training data, the structures outside the FOV can be reconstructed better. This is demonstrated by the results in Fig. 11, where the Pix2pixGAN-based PnP results from different training data sizes are displayed. Fig. 11 shows that the structures inside the FOV have no observable difference in the PnP results with different training data sizes. But with more training data, the structures outside the FOV (e.g., the bottom soft tissue boundary) are reconstructed better.

The existence of adversarial examples is the third factor to influence the robustness of deep learning. In our previous work [31], we have demonstrated that Poisson noise is actually one type of adversarial perturbations, which has low magnitude but can change the presence of a lesion in the deep learning output. In this work, we also observe false negative lesion cases (Figs. 7 and 8) in the noisy case. By bringing back the consistency with measured data, our PnP method is able to correct false structures inside the FOV. In addition, even for other adversarial examples which can significantly change the deep learning output image, our PnP method can still guarantee the structures inside the FOV to be faithful, although structures outside the FOV are not reliable. This is demonstrated by the results displayed in Fig. 10. These observations imply that PnP is able to correct false structures inside the FOV caused by adversarial perturbations.

In the noise-free case (Fig. 6), we perform extra experiments for TV regularization because we want to investigate whether TV can further improve incorrect structures outside the FOV like it does in other CT reconstruction tasks from insufficient data [13]. However, no significant improvement is observed, potentially because the angular coverage decreases drastically when the distance to the isocenter increases. In addition, it also helps us to analyze the influence of each regularization term in the overall objective functions (Eqn. (5) and Eqn. (12)). With the extra experiments, we can conclude that the image quality improvement regarding truncation artifacts is from the learned prior regularization. And the function of the TV regularization is mainly for noise reduction and reduction of streak artifacts in the clinical case.

Pix2pixGAN has significant advantage over FBPConvNet, especially in the noisy case, as demonstrated by Figs. 4, 7 and 8. In the noisy case, the results of FBPConvNet look very blurry and unrealistic, although it is trained on noisy data. It is because the ℓ_2 loss is used to train FBPConvNet. It is well known that the ℓ_2 loss encourages finding pixel averages of plausible solutions that are typically overly smoothed [66], [67]. Although the loss is minimized, the generated images typically have poor perceptual quality from a perspective of human observers [68]. In this work, Pix2pixGAN also applies the ℓ_2 loss for training a generative model to minimize the difference between the output image and the target image. However, an additional discriminator is simultaneously trained that tries to classify whether the output image is real or fake. Blurry images will not be tolerated by the discriminator since they look obviously fake. Because Pix2pixGAN learns a combined loss that adapts to the data, it is able to deal with multiple tasks (denoising and truncation correction) better than FBPConvNet [60]. Nevertheless, in Fig. 7, FBPConvNet is able to predict the outlines of the body and the major organs, which already provides good prior for extrapolation. The anatomical details can be recovered to some degree by the third step of our PnP method.

Figs. 4-6 demonstrate that the proposed PnP method is capable of correcting false positive lesions. For true lesions, deep learning solely has a risk to miss true lesions, which are the false negative cases displayed in Fig. 7(j) and Fig. 8(d). Our PnP method is able to correct such false negative cases

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and reconstruct the true lesions, as demonstrated in Fig. 7(k)-(l) and Figs. 8(e)-(f). In addition, PnP is able to improve true positive cases of deep learning results. Although deep learning methods are able to reconstruct true lesions, the location and area of the lesion might be inaccurate. For example, the lesion reconstructed by Pix2pixGAN displayed in Fig. 8(d) has much smaller area. PnP is able to improve the true lesion region, as demonstrated by Figs. 8(b)-(c). In other words, PnP can ensure the reliability of structures inside the FOV. However, the limitation of PnP lies at the structures outside the FOV. These structures highly rely on learned images and the improvement of incorrect structures outside the FOV is not guaranteed, as indicated by Fig. 10.

The main purposes of our proposed PnP truncation correction algorithm lies at the following aspects: a) Ensure all reconstructed structures inside the FOV are reliable; b) Have a natural transition region at the FOV boundary; c) Restore structures outside the FOV as much as possible, which relies on learned images. For the first point, measured projection data typically contain noise or errors caused by various physical effects such as beam-hardening and scatter, introducing artifacts. Such artifacts can be preserved by our PnP algorithm with a naive analytic reconstruction. For example, streak artifacts indicated by the arrow in Fig. 11(a) are preserved in Fig. 11(e). However, when the artifacts can be reduced by existing conventional reconstruction algorithms, such algorithms can be plugged in to reduce artifacts. Since the properties of such conventional algorithms are known, the reconstructed structures inside the FOV are still reliable and interpretable. For example, iterative reconstruction has advantages over analytic reconstruction in reducing artifacts caused by cone-beam angles, beam-hardening and scattering [69], [70]. Hence, the streak artifacts in Fig. 11(a) are reduced in our PnP reconstruction using iterative reconstruction in Figs. 11(f) and (l).

As a hybrid method combining deep learning with conventional reconstruction, PnP preserves their respective advantages: a) PnP is as robust as conventional reconstruction methods for structures inside the FOV, since both of them keep the data fidelity with measured data. b) PnP is as effective as deep learning to reconstruct structures outside the FOV. Thanks to the information provided by the learned image prior for the missing data, PnP preserves or even improves anatomical structures outside the FOV. c) PnP is relatively efficient. For FBP-based PnP, it requires slice-wise deep learning inference, projection domain data extrapolation and one FBP reconstruction step. For iterative reconstruction based PnP, 10 iterations only (Fig. 9) are sufficient to converge using the learned images as initialization, while conventional iterative reconstruction typically requires much more iterations.

V. Conclusion

In this work, two example neural networks, FBPConvNet and Pix2pixGAN, are applied for CT truncation correction. Pix2pixGAN achieves better performance than FBPConvNet both in the noise-free and noisy cases. The robustness of FBPConvNet and Pix2pixGAN is investigated by exemplifications of false positive and false negative lesion cases. To improve deep learning reconstruction, the PnP method is proposed combining the advantages of deep learning and conventional reconstruction. With PnP, a final image is reconstructed by conventional reconstruction from data extrapolated from learned prior images along with measured data. In such a combination, the high representation power of deep learning and the high robustness as well as interpretability of conventional reconstruction are integrated together in the proposed PnP method.

Disclaimer: The concepts and information presented in this paper are based on research and are not commercially available.

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